Computer Science & IT



Data Structure & Programming

Tree

Lecture No. 06



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Recap of Previous Lecture



- Topic
- Topic
- Topic
- Topic
- Topic

BST, Search,

Min, Max

Counting

$O(h)$ <

- BST $O(n)$
- balanced $O(\log_2 n)$

Slide

Topics to be Covered



- Topic — BST Deletion
- Topic — AVL Tree.
- Topic
- Topic
- Topic

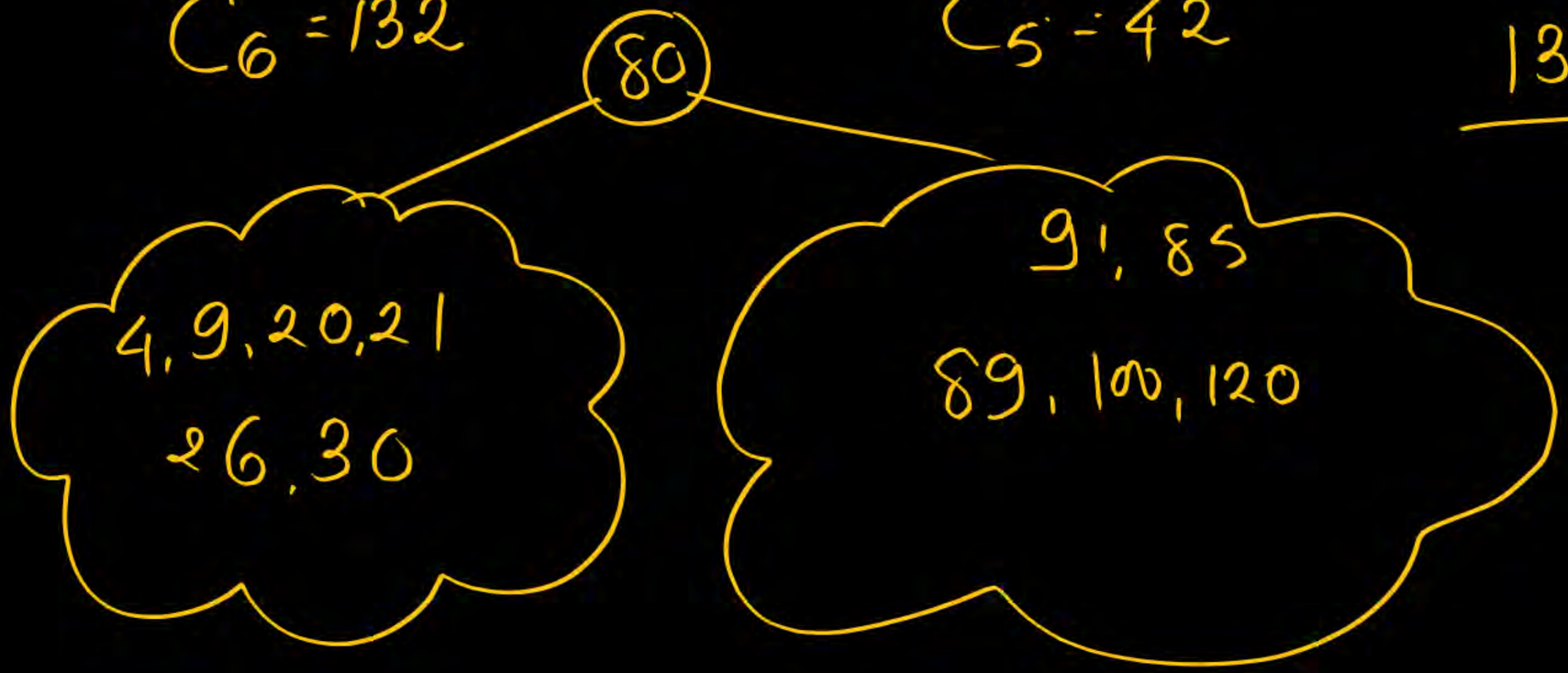
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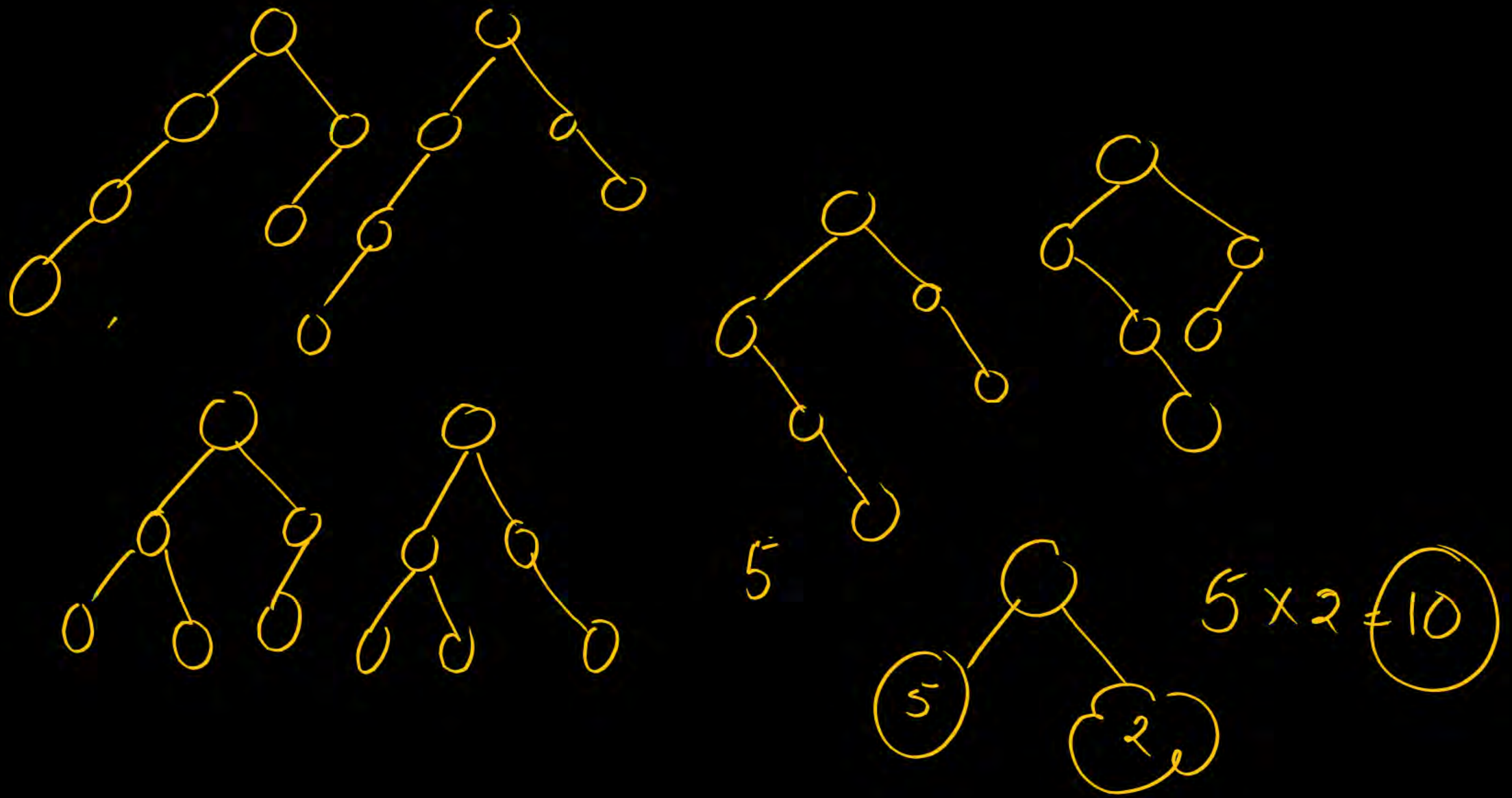
4, 9, 20, 21, 26, 30, 91, 80, 85, 89, 100, 120

$C_6 = 132$

$C_5 = 42$

$132 \times 42 = 5544$




5
5
2
5 x 2 ≠ 10



Topic : Tree

BST delete
AVL tree insertion & rotation

Slide

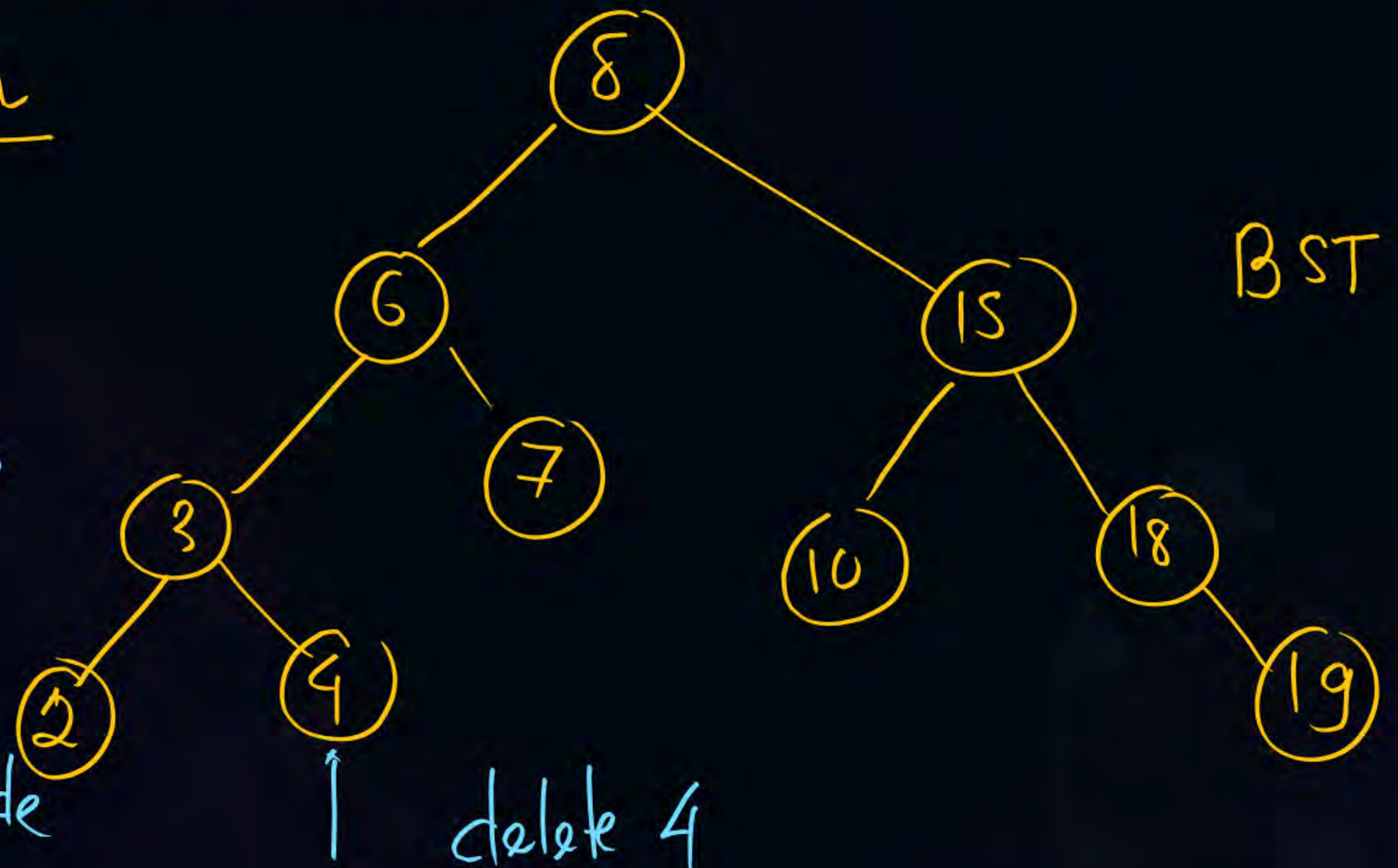


Topic : Binary Search Tree

Deletion

Case (I)

if we want to delete a node which is leaf Node



BST

Slide



Topic : Binary Search Tree

Deletion

Case(I)

if we want to delete a node which is leaf Node



BST

Slide



Topic : Binary Search Tree

Deletion

Case-II If you want to delete a node x having 1 children



BST

Slide



Topic : Binary Search Tree

Delehon



x is a Node

P[x] parent

Node of x

Slide



Topic : Binary Search Tree

Deletion

Case II : If we delete a Node having 1 children then p[x] (parent of x) becomes parent of children x



x is a Node

P[x] parent Node of x

Slide



Topic : Binary Search Tree

Deletion

Case II : If we delete a Node having 1 children then p[x] (parent of x] becomes parent of children x



x is a Node

P[x] parent Node of x

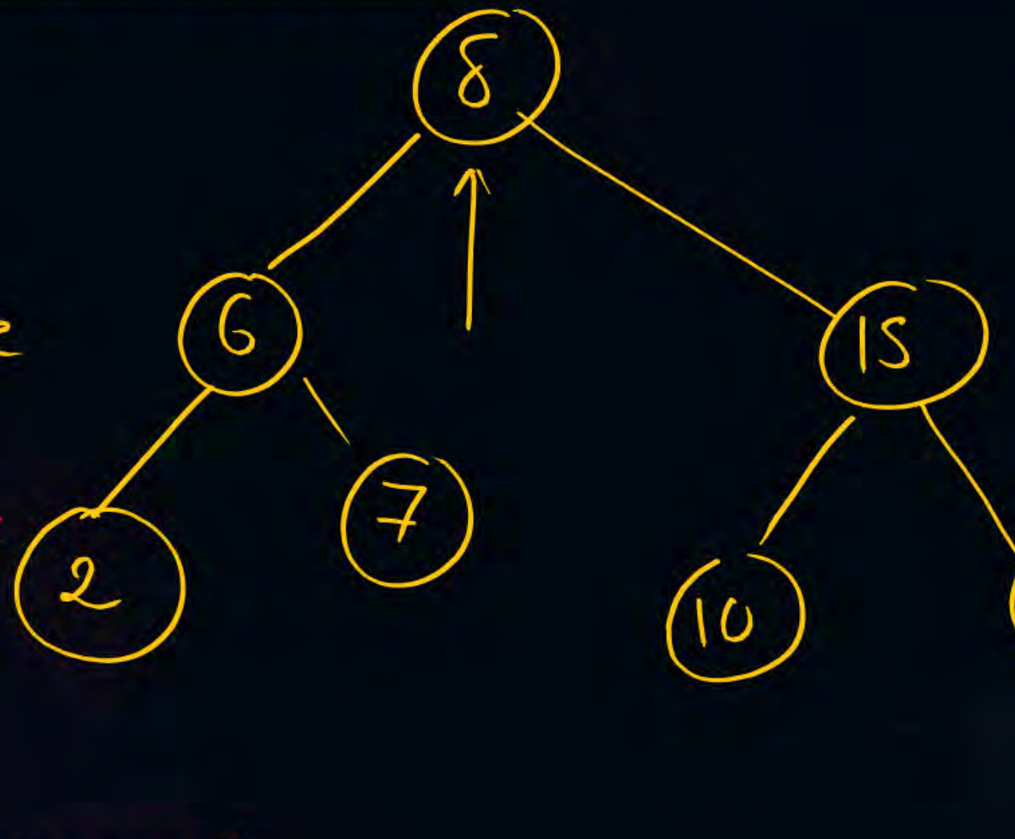
Slide



Topic : Binary Search Tree

(Case III)

If we want to delete a Node with 2 children the its replaced by either Inorder predecessor or Inorder Successor



x is a Node

P[x] parent Node of x



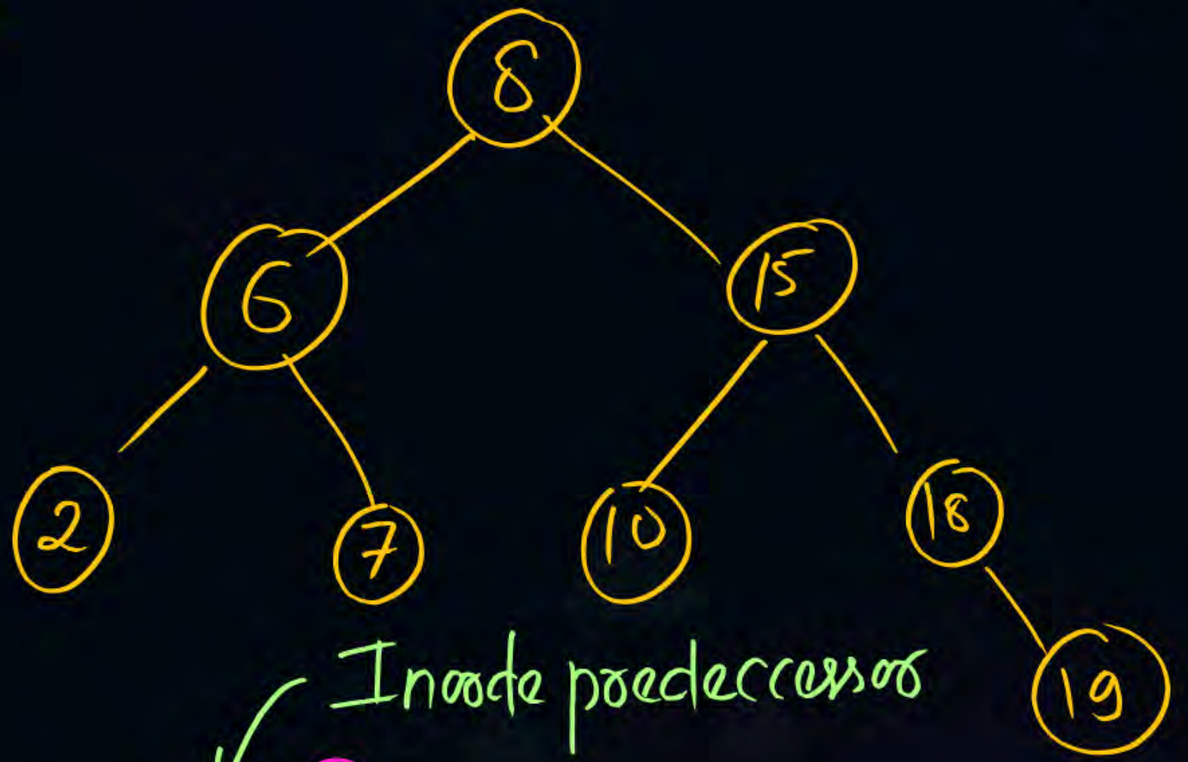
Topic : Binary Search Tree

for a given Node x of BST,

Inorder **predecessor** is the node that encounter **before** Node x in Inorder traversal.

Predecessor of min &

Successor of max does not exists.



Slide



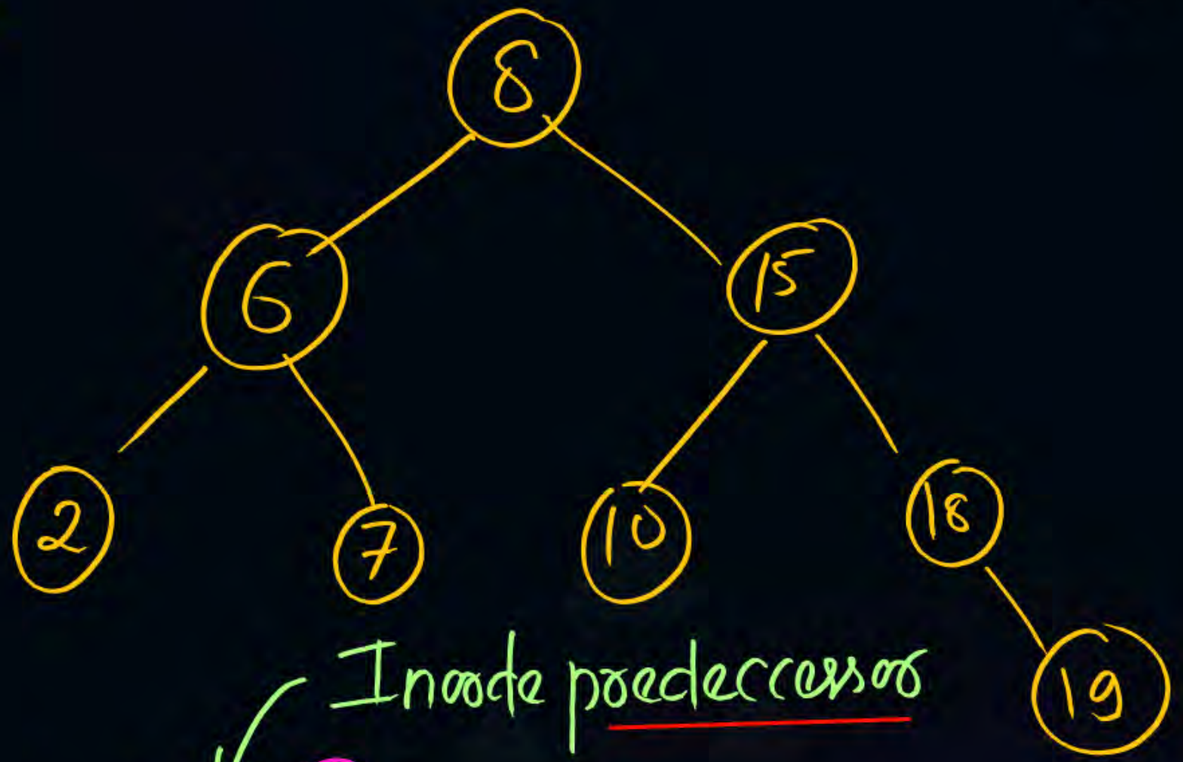
Topic : Binary Search Tree

for a given Node x of BST,

Inorder **Successor** is the node that encounter **after** Node x in Inorder traversal.

Predecessor of min &

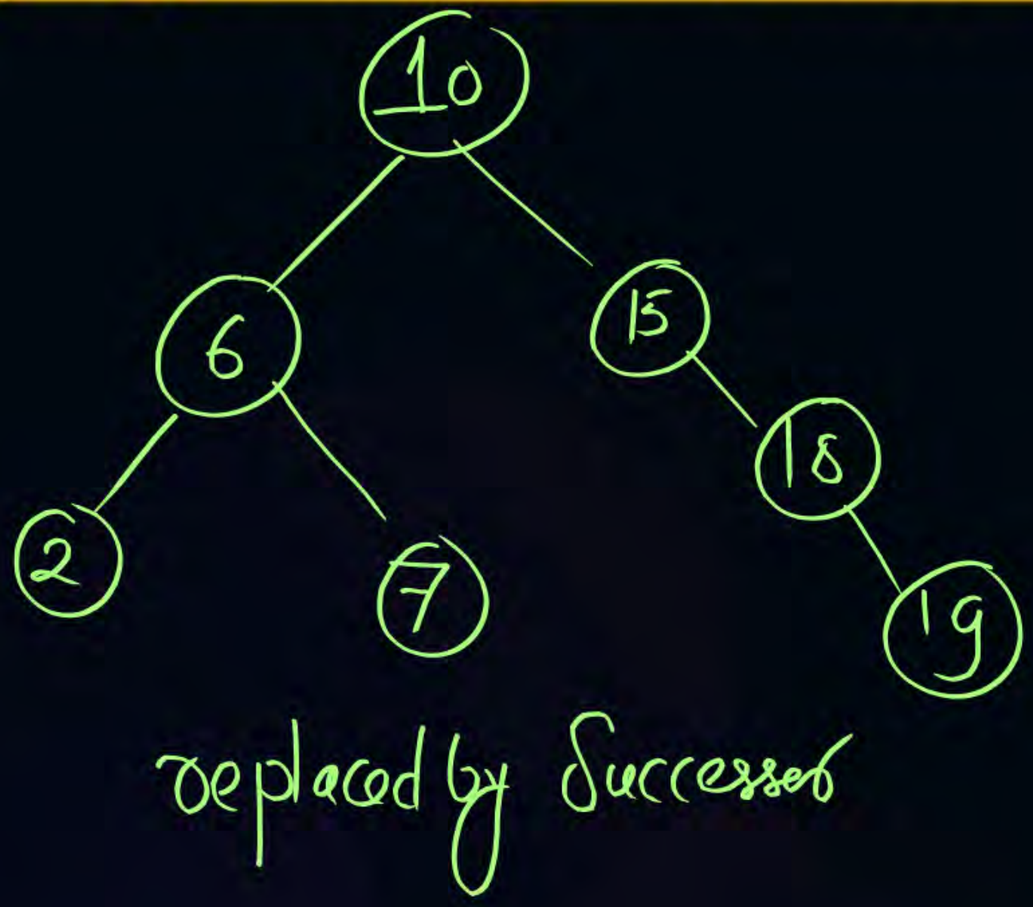
Successor of max does not exists.



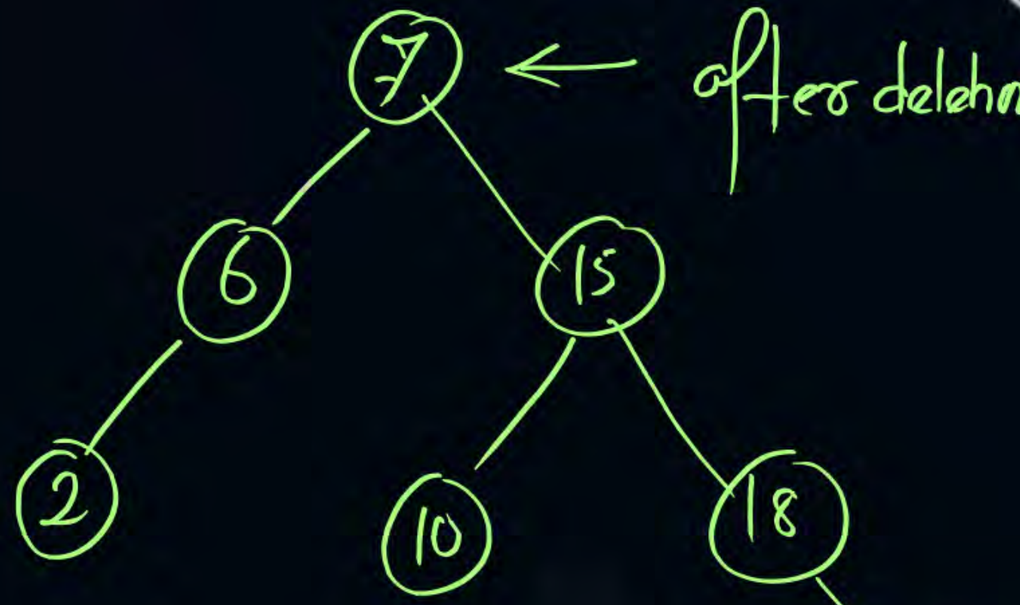
Slide



Topic : Binary Search Tree



replaced by Successor



Replaced by predecessor

Slide



Topic : AVL Tree

BST — Insert
Delete
Min
Max
Search

} Height dependent

worst case O(n) Linear time



Topic : AVL Tree

AVL Tree : Height balanced tree

A BST is an AVL tree if difference of height in Left subtree and right subtree is 0, 1 or -1.

Balancing factor

Slide

Topic : AVL Tree



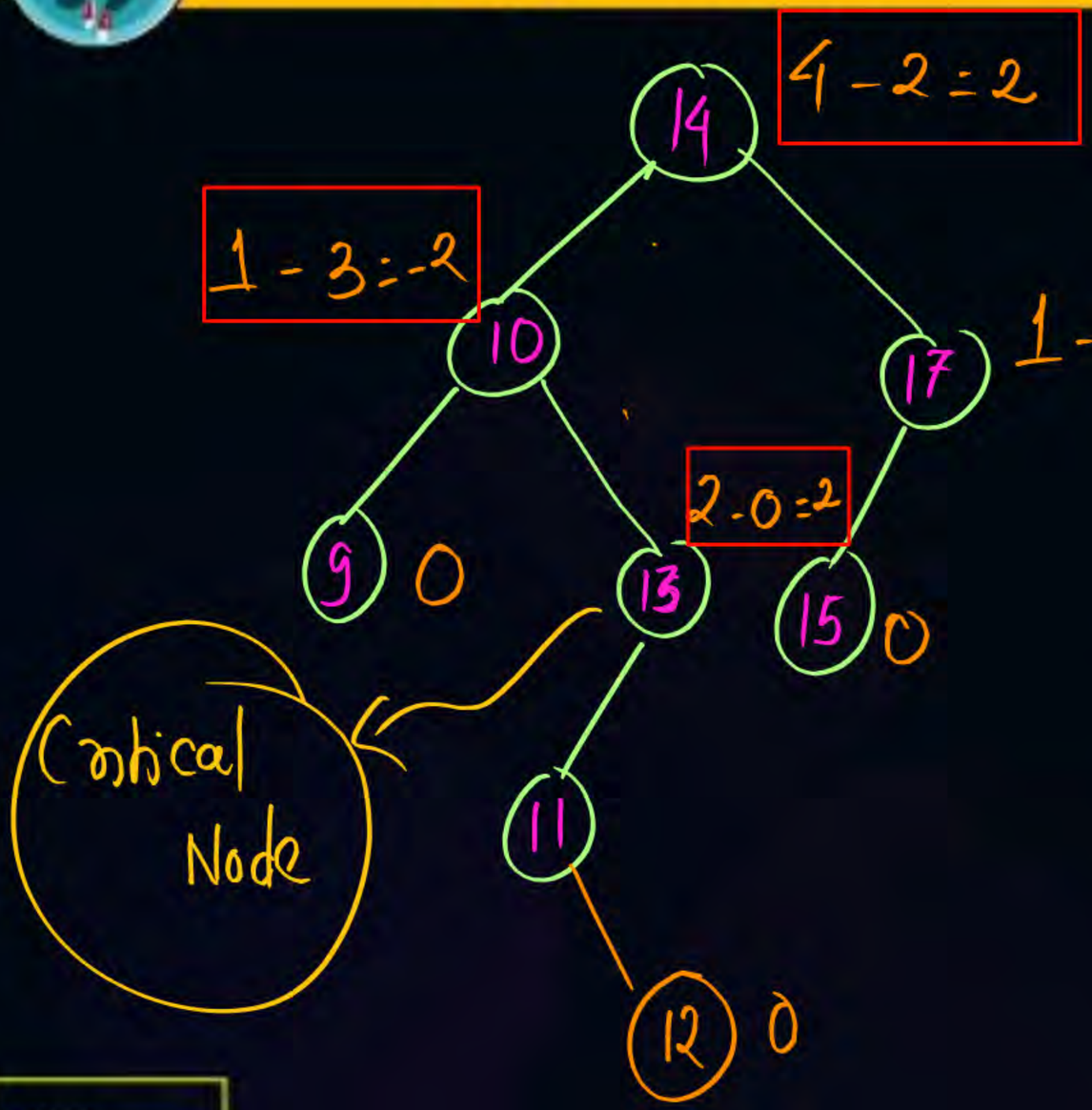
Is this a AVL tree

Left Height - Right Height

Right Height - Left Height

Slide

Topic : AVL Tree



Not AVL Tree

Critical Node

After insertion balancing factor violated by first Node in Leaf to Root path is critical Node.

Slide



Topic : AVL Tree

Balancing factor 0, 1, -1 changes due to Insertion in AVL tree.

Slide



Topic : AVL Tree

if critical Node is present then we Need to perform Rotation to balance it:

* Single Rotation
  - LL Rotation
  - RR Rotation
* Double Rotation
  - LR Rotation
  - RL Rotation

Slide



Topic : AVL Tree

Single Rotation LL Rotation.

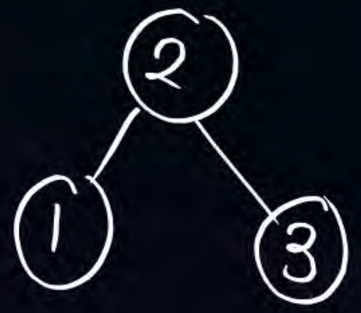
Not AVL tree

Insert 3, 2, 1



← critical

Right Rotation



with respect to critical

Node Newly inserted

Node in left subtree

Left direction (child)

Slide



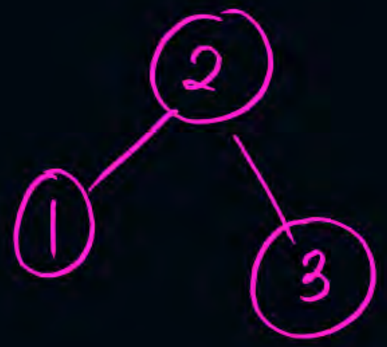
Topic : AVL Tree

Insert 1, 2, 3



RR Rotation

one left Rotation will fix it.



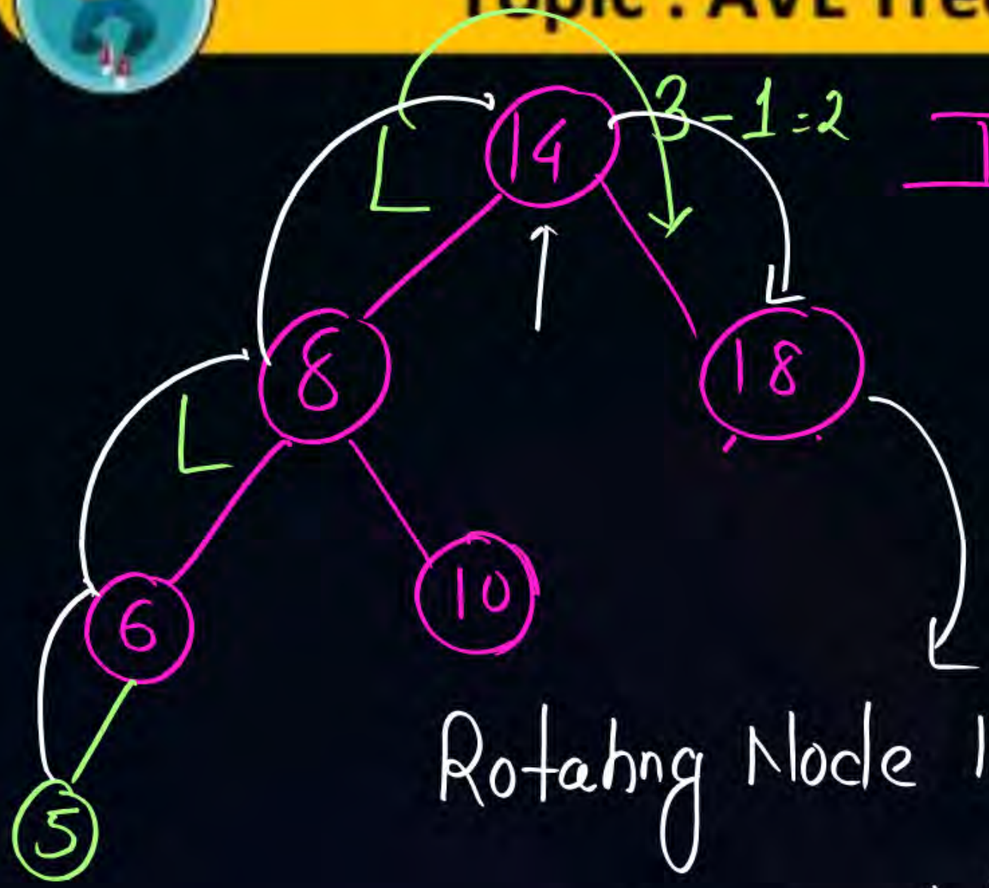
With respect to critical

Node newly inserted Node is in Right Subtree2 Right direction.

Slide



Topic : AVL Tree



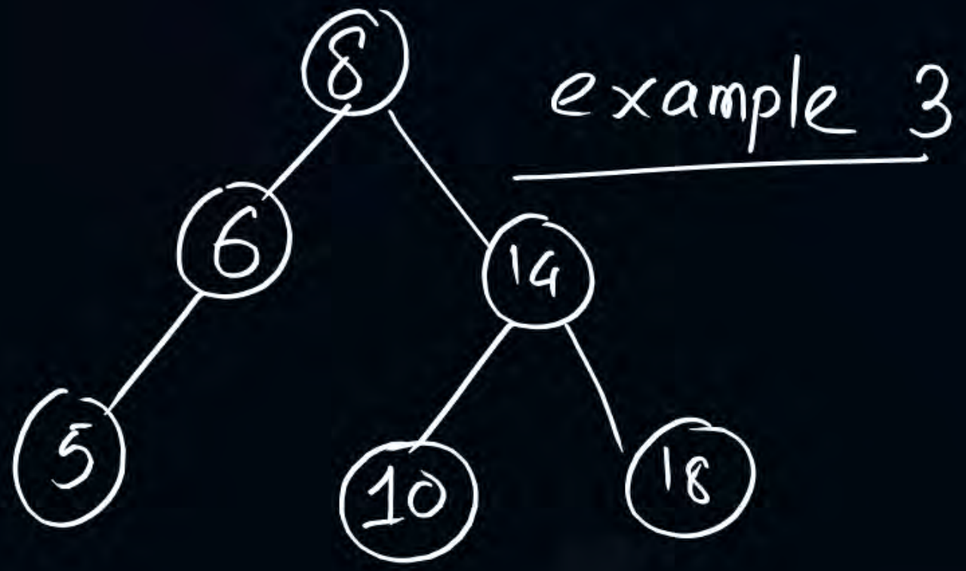
Insert 5

Rotating Node 14

10 is in left subtree of 14

after Rotation 10 will be Left child of (14)

example 3



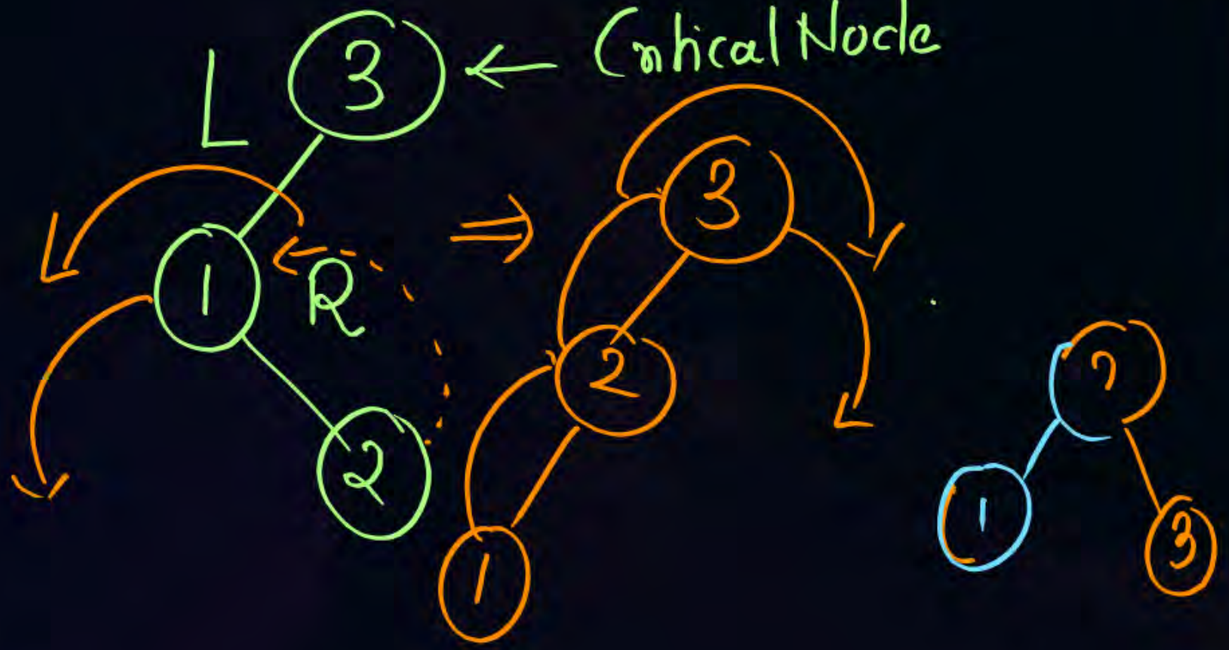
Slide



Topic : AVL Tree

Double Rotation LR Rotation

Insert 3, 1, 2



LR

- Left child of critical Node Rotated left
- Critical Node Rotated Right

Slide



Topic : AVL Tree

With Respect to Critical Node newly inserted

Node is in Left Subtree & Right Direction

Slide

Topic : AVL Tree

Insert 1, 3, 2



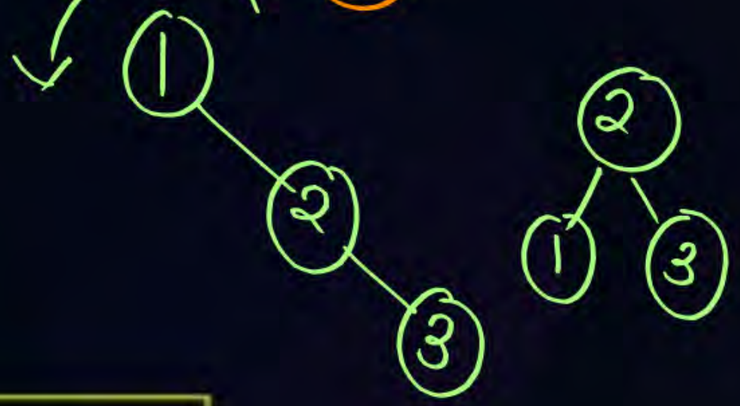
← critical Node

from critical Node to newly inserted Node we Need to go Right Subtree & left direction

R L

→ Right child of critical Node Rotated Right

→ Critical Node Rotated left.



Slide



Topic : AVL Tree

3, 2, 1



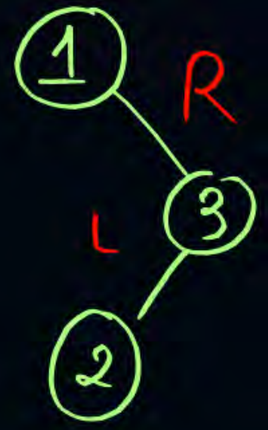
Right Rotation

1, 2, 3



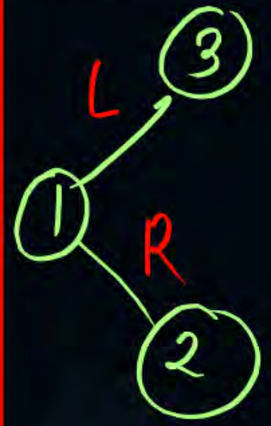
Left Rotation

1, 3, 2



RL Rotation

3, 1, 2



LR Rotation

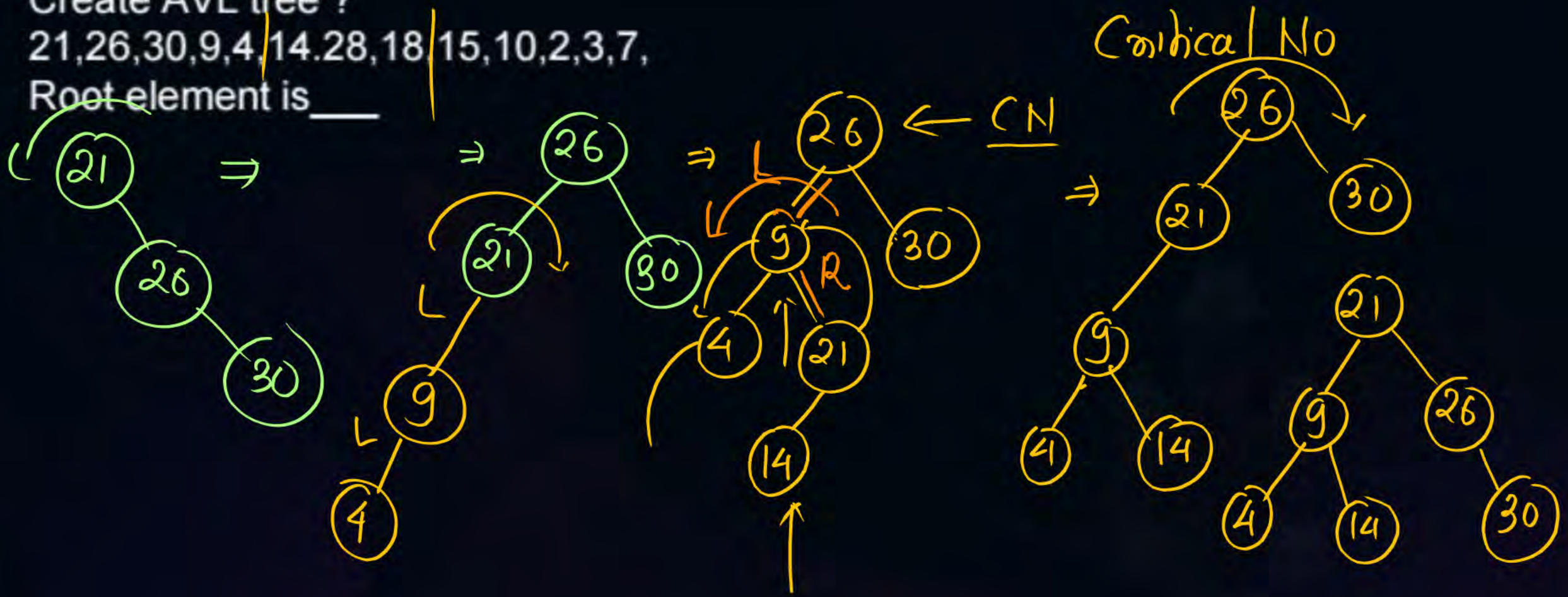
Slide

Topic : AVL Tree

Create AVL tree ?

21,26,30,9,4,14,28,18,15,10,2,3,7,

Root element is___



Slide

Topic : AVL Tree

Create AVL tree ?

21,26,30,9,4,14,28,18,15,10,2,3,7,

Root element is___

find Root of tree.



Slide

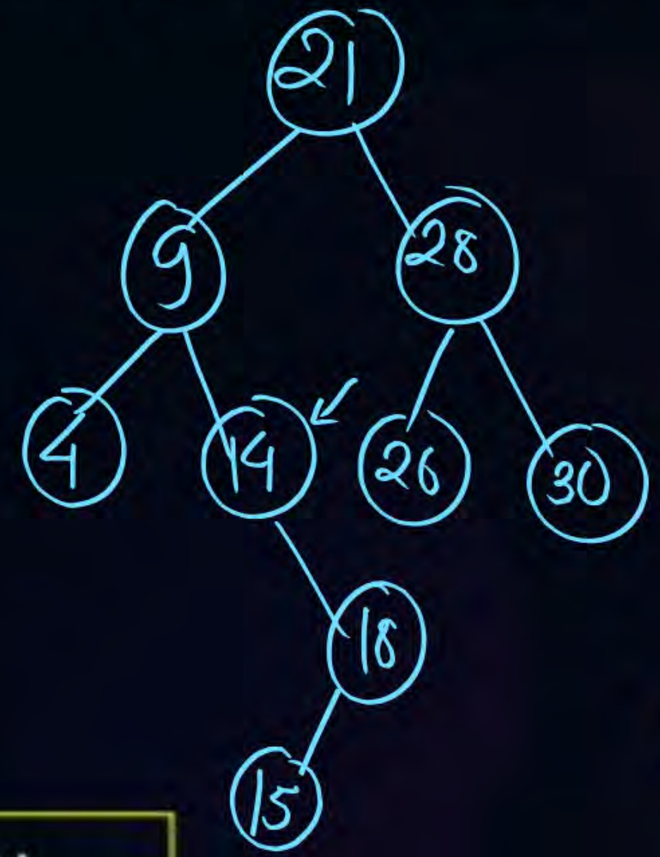


Topic : AVL Tree

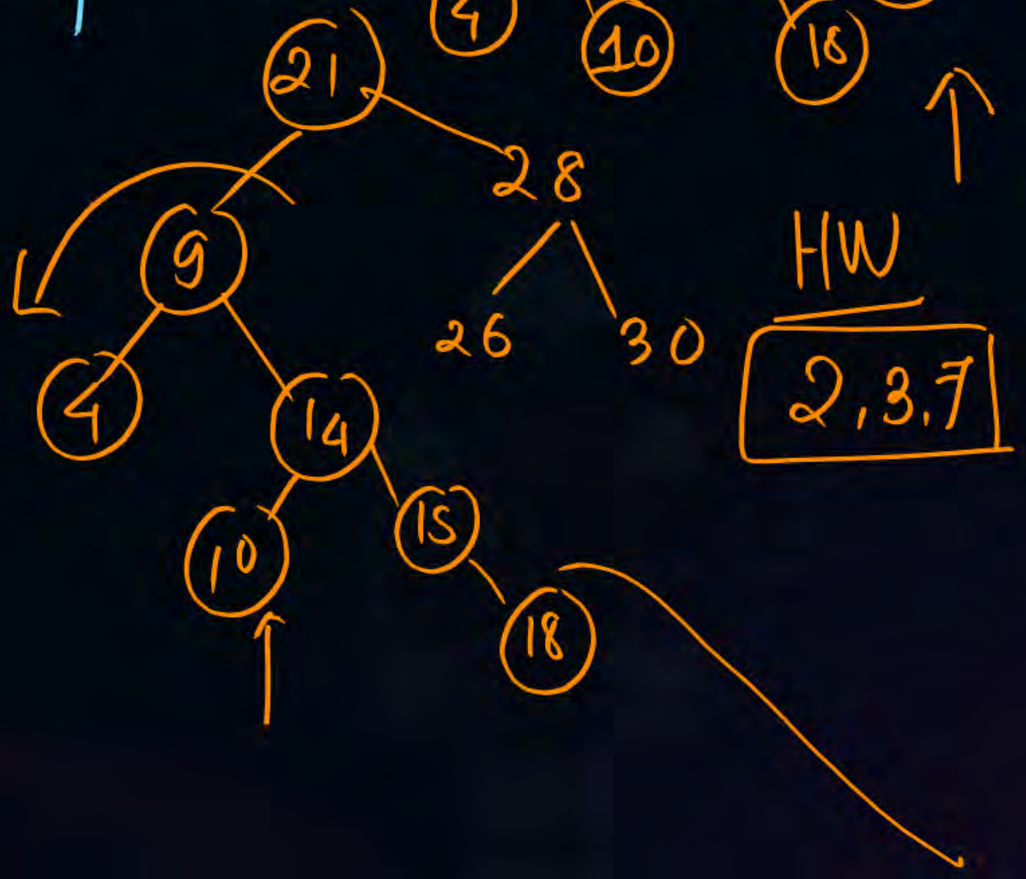
Create AVL tree ?

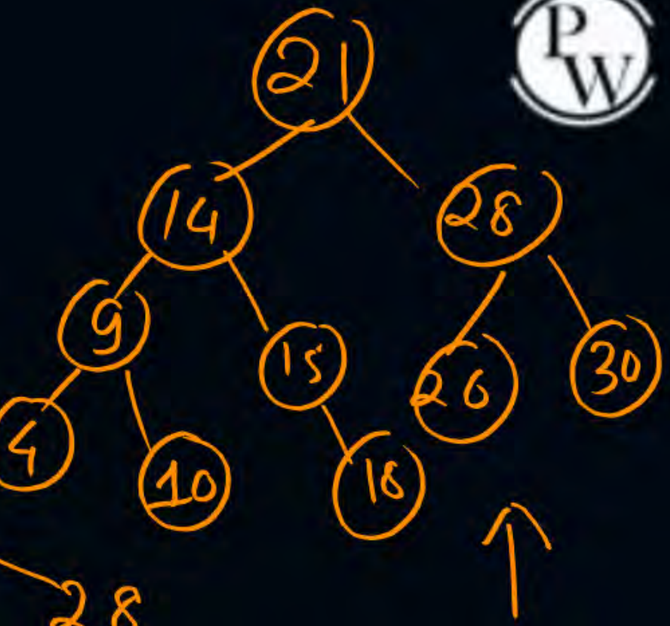
21,26,30,9,4,14.28,18,15,10,2,3,7,

Root element is___





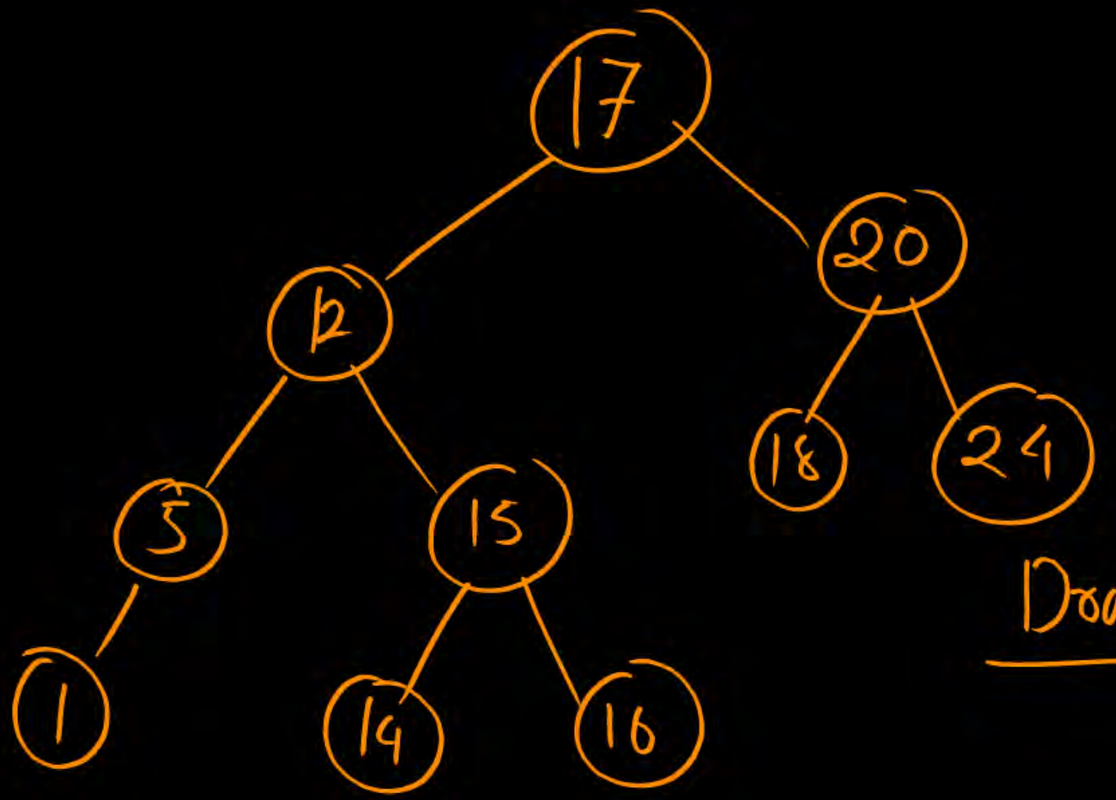




HW

2,3,7

Slide



Insert 13

Draw final tree



2 mins Summary

Topic BST Delek

Topic AVL tree

Topic AVL Rotahon

Topic practice problem

Topic

Slide

THANK - YOU

Slide